Kernix 0.9.0 / Performance brief

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

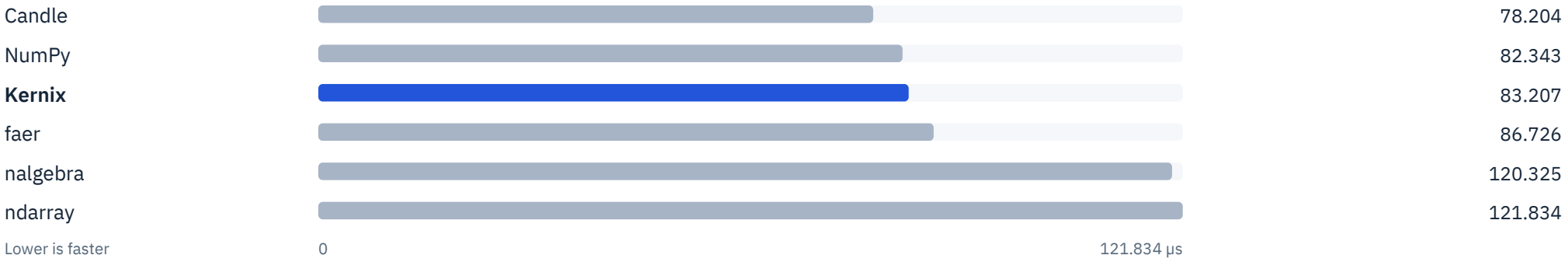
31 workloads

108 observations

Source precision retained

Matrix multiplication

128 × 128 · 128 × 128 · f64 · 1 worker · allocated output · self product



How to read this report

Allocation-inclusive API latency unless marked reused output. Peer medians are observations.

Charts round to 6 significant digits (≈). The exact-value appendix preserves all recorded digits.

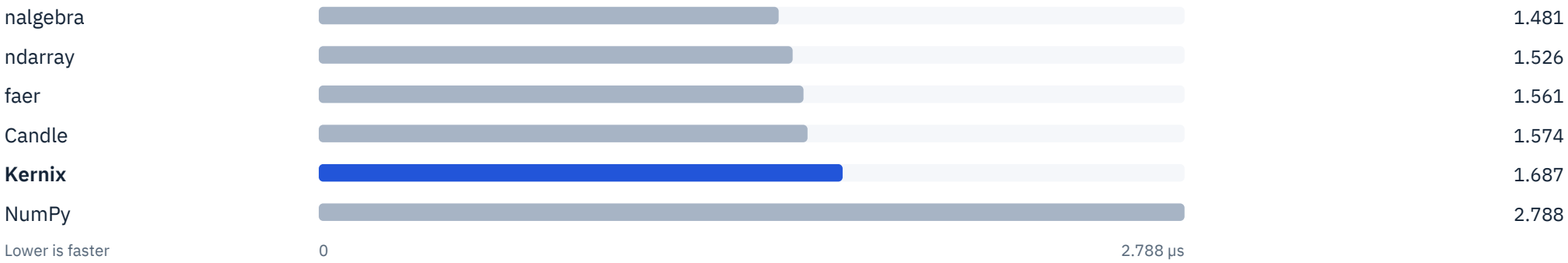
Kernix 0.9.0 / Matrix multiplication

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

Matrix multiplication

Per-call median

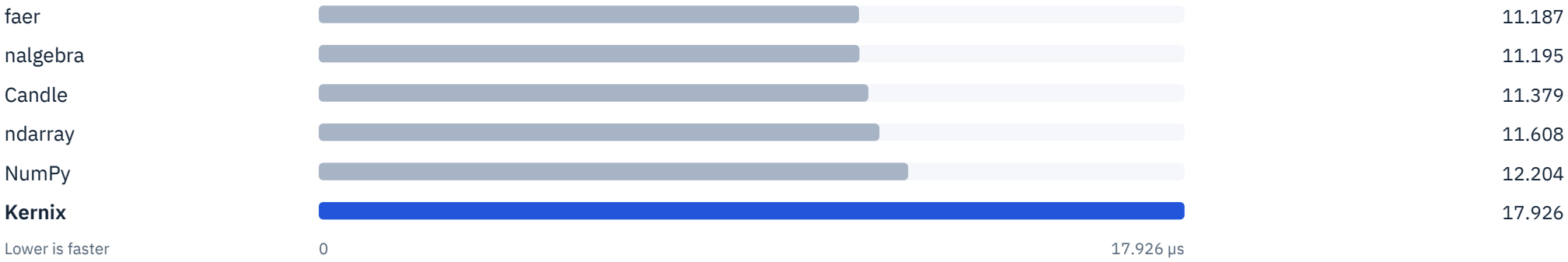
32 × 32 · 32 × 32 · f64 · 1 worker · allocated output · self product



Matrix multiplication

Per-call median

64 × 64 · 64 × 64 · f64 · 1 worker · allocated output · self product



Kernix 0.9.0 / Matrix multiplication

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

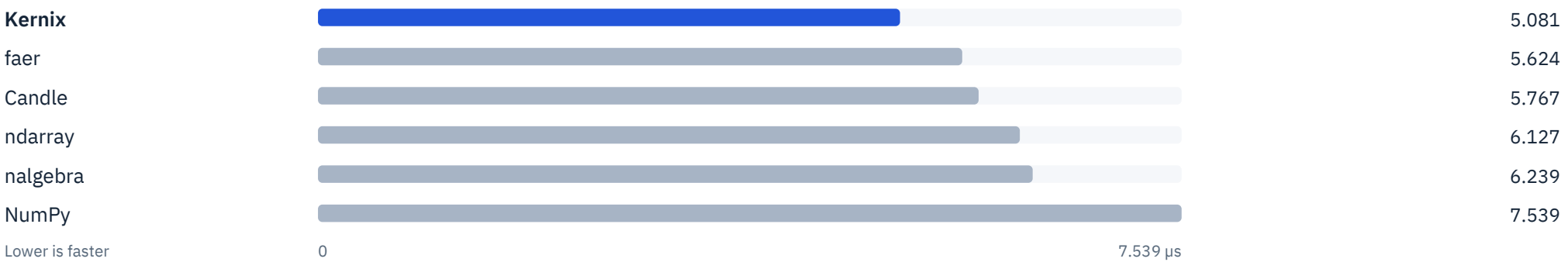
Matrix multiplication

256 × 256 · 256 × 256 · f64 · 1 worker · allocated output · self product



Matrix multiplication

64 × 64 · 64 × 64 · f32 · 1 worker · allocated output · self product



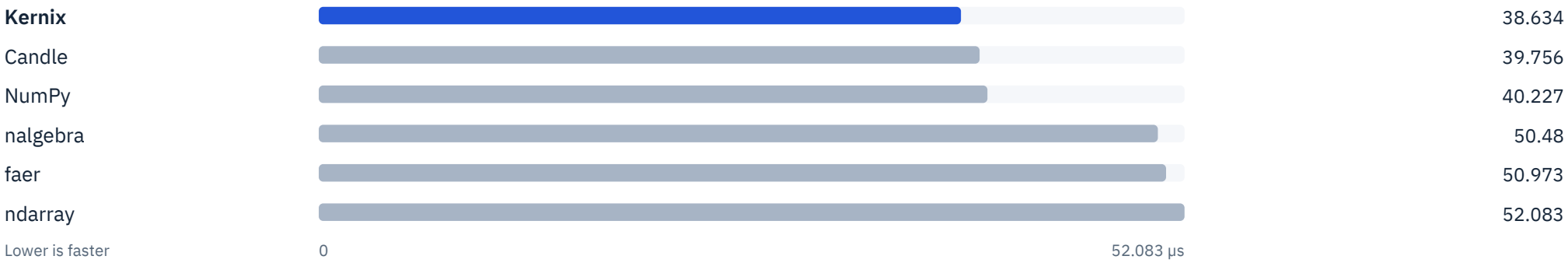
Kernix 0.9.0 / Matrix multiplication

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

Matrix multiplication

Per-call median

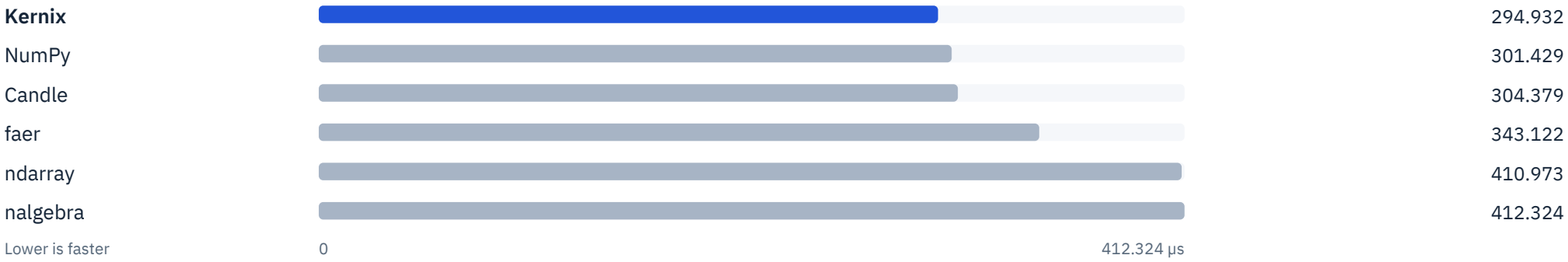
128 × 128 · 128 × 128 · f32 · 1 worker · allocated output · self product



Matrix multiplication

Per-call median

256 × 256 · 256 × 256 · f32 · 1 worker · allocated output · self product

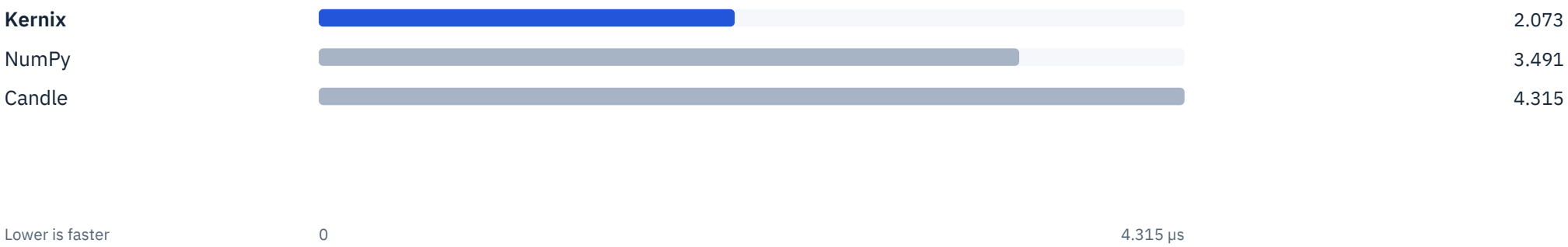


Kernix 0.9.0 / Matrix multiplication

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

Batched multiplication

4 × 16 × 32 · 1 × 32 × 16 · f64 · 1 worker · allocated output · broadcast batches



Kernix 0.9.0 / Explicit parallel execution

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

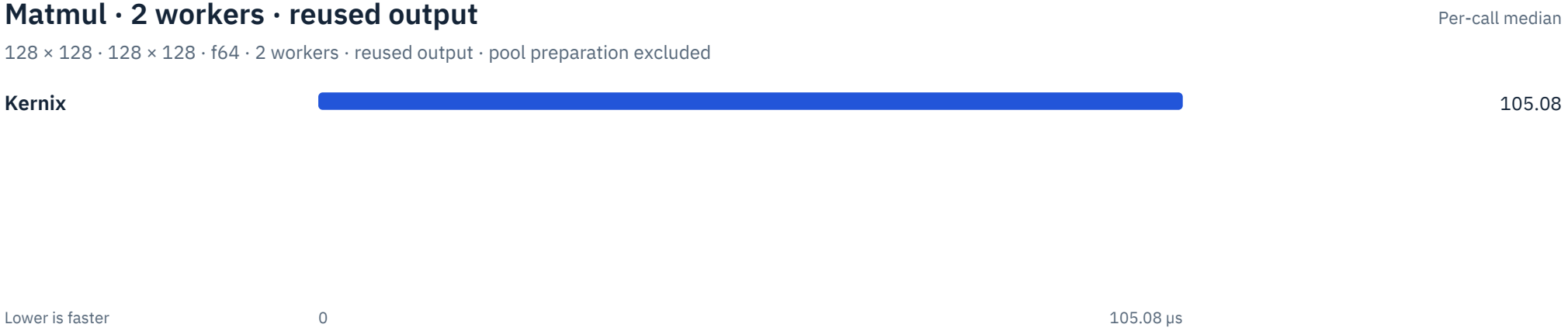
Matmul · 2 workers · reused output

64 × 64 · 64 × 64 · f64 · 2 workers · reused output · pool preparation excluded



Matmul · 2 workers · reused output

128 × 128 · 128 × 128 · f64 · 2 workers · reused output · pool preparation excluded



Kernix 0.9.0 / Explicit parallel execution

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

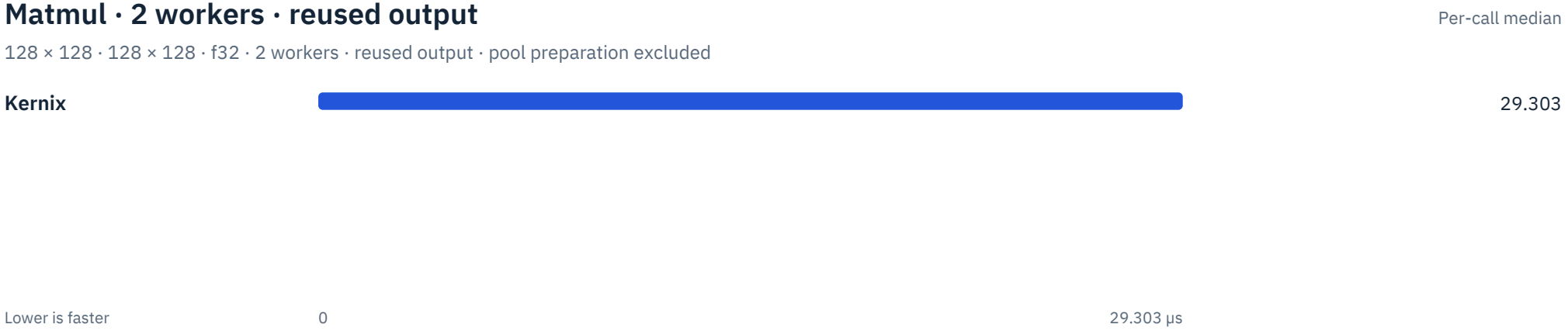
Matmul · 2 workers · reused output

256 × 256 · 256 × 256 · f64 · 2 workers · reused output · pool preparation excluded



Matmul · 2 workers · reused output

128 × 128 · 128 × 128 · f32 · 2 workers · reused output · pool preparation excluded



Kernix 0.9.0 / Decompositions & solvers

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

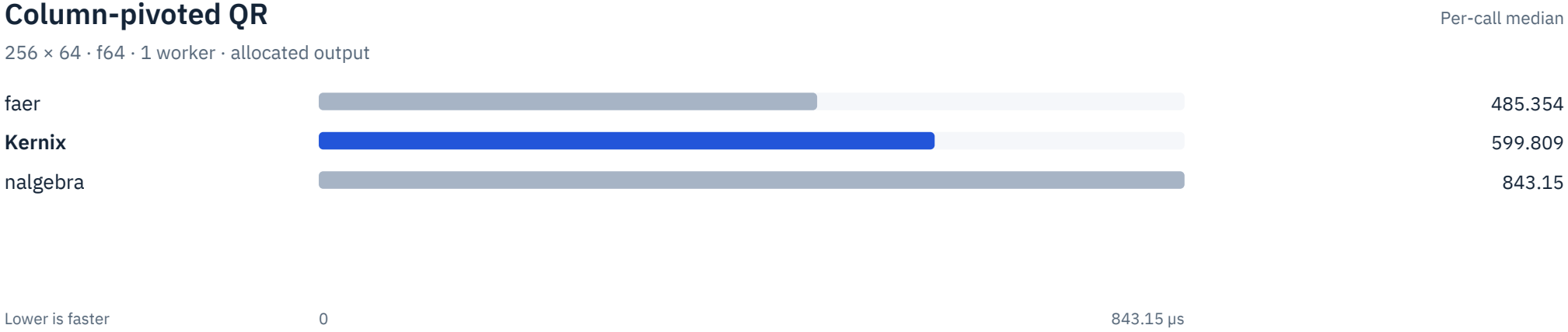
Column-pivoted QR

96 × 32 · f64 · 1 worker · allocated output



Column-pivoted QR

256 × 64 · f64 · 1 worker · allocated output



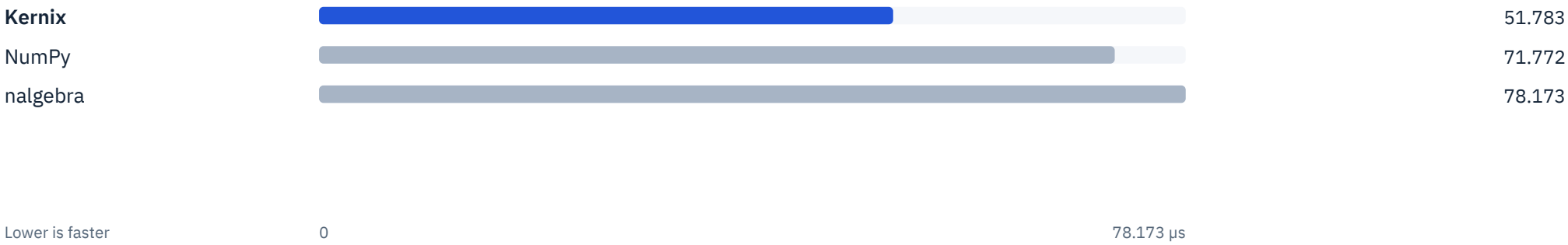
Kernix 0.9.0 / Decompositions & solvers

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

Minimum-norm least squares

Per-call median

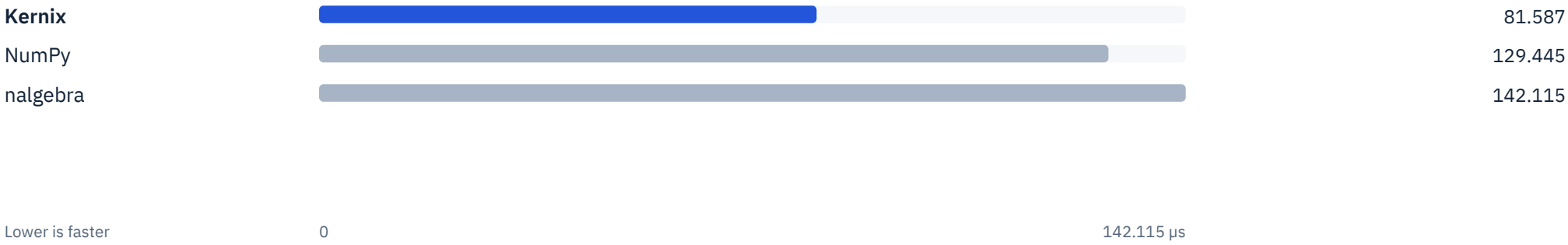
64 × 32, rank 16, 4 RHS · f64 · 1 worker · 4 RHS · factorization included · algorithms differ



Minimum-norm least squares

Per-call median

32 × 64, rank 32, 4 RHS · f64 · 1 worker · 4 RHS · factorization included · algorithms differ

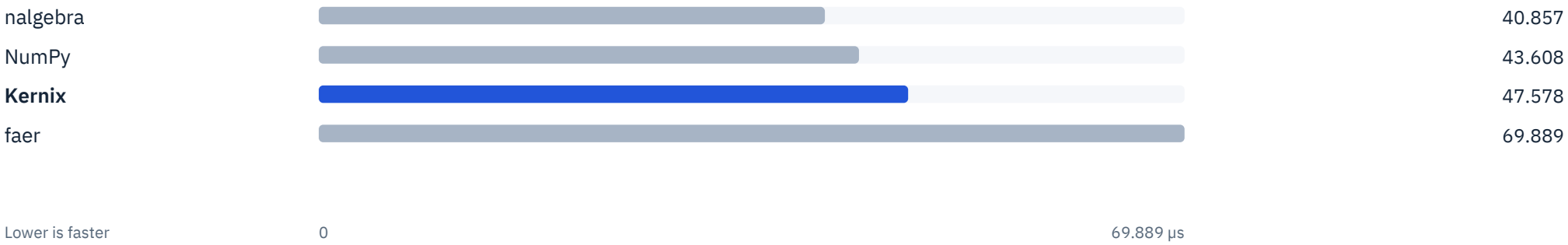


Kernix 0.9.0 / Decompositions & solvers

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

Thin QR

96 × 32 · f64 · 1 worker · allocated output



Thin QR

256 × 64 · f64 · 1 worker · allocated output



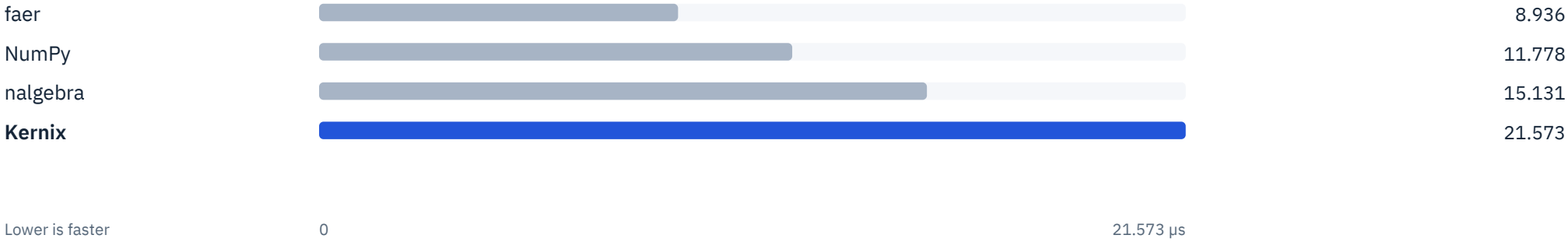
Kernix 0.9.0 / Decompositions & solvers

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

Cholesky factorization

Per-call median

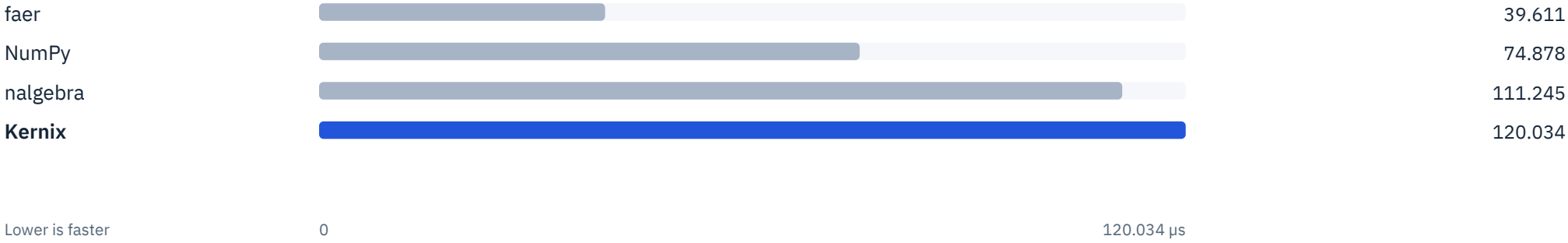
64 × 64 · f64 · 1 worker · allocated output



Cholesky factorization

Per-call median

128 × 128 · f64 · 1 worker · allocated output



Kernix 0.9.0 / Decompositions & solvers

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

Cholesky factorization

256 × 256 · f64 · 1 worker · allocated output

Per-call median



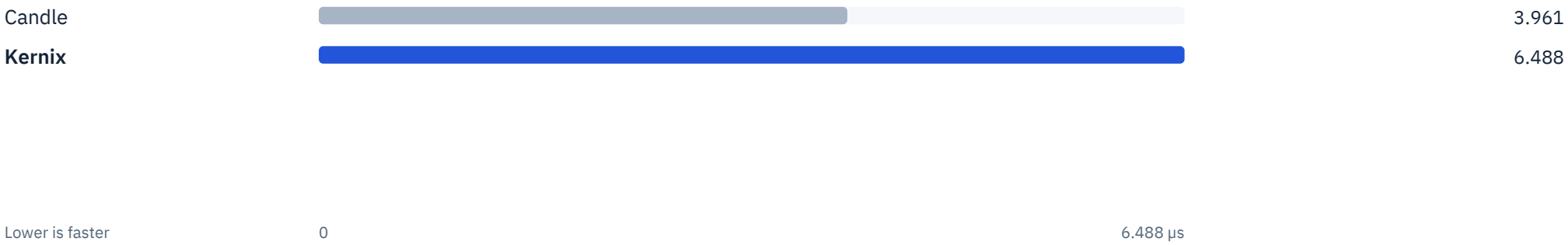
Kernix 0.9.0 / Training & autodiff

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

Shared-variable backward

Per-call median

sum(X * X), 64 × 32; prepared graph · f64 · 1 worker · prepared graph · fresh gradients



Record + backward

Per-call median

sum(A · W), 32 × 16 · 16 × 8 · f64 · 1 worker · allocated output · self product



Kernix 0.9.0 / Training & autodiff

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

Model inference

Per-call median

batched normalized/sliced MLP · f64 · 1 worker · preparation as documented



Model recording + forward

Per-call median

batched normalized/sliced MLP · f64 · 1 worker · preparation as documented



Kernix 0.9.0 / Training & autodiff

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

Model backward

Per-call median

prepared graph → fresh gradients · f64 · 1 worker · prepared graph · fresh gradients



Parameter update

Per-call median

transactional SGD, fixed gradients · f64 · 1 worker · preparation as documented



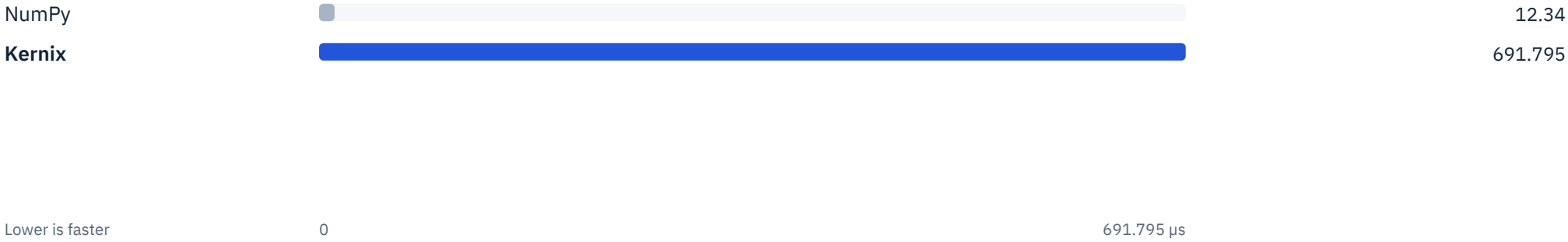
Kernix 0.9.0 / Tensor contractions

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

Coalesced tensor contraction

16 × 32 × 8 · 32 × 8 × 16 · f64 · 1 worker · allocated output

Per-call median



Kernix 0.9.0 / Statistics & activations

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

Row softmax

256 × 64 · f64 · 1 worker · allocated output

Per-call median



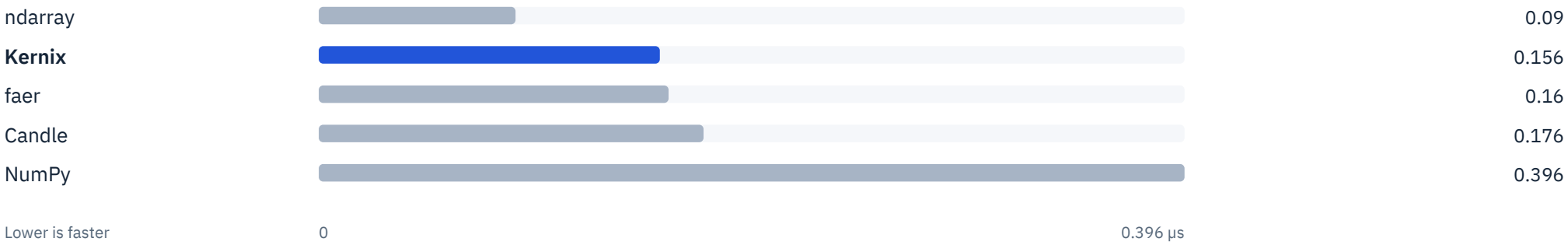
Kernix 0.9.0 / Tensor construction

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

Create zeros

Per-call median

64 × 64 · f32 · 1 worker · allocated output



Create ones

Per-call median

64 × 64 · f32 · 1 worker · allocated output



Recorded precision / exact latency table

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Matrix multiplication / faer 32 × 32 · 32 × 32 · f64	1561	1.561
Matrix multiplication / Candle 32 × 32 · 32 × 32 · f64	1574	1.574
Matrix multiplication / NumPy 32 × 32 · 32 × 32 · f64	2788	2.788
Matrix multiplication / Kernix 32 × 32 · 32 × 32 · f64	1687	1.687
Matrix multiplication / ndarray 32 × 32 · 32 × 32 · f64	1526	1.526
Matrix multiplication / nalgebra 32 × 32 · 32 × 32 · f64	1481	1.481
Matrix multiplication / ndarray 64 × 64 · 64 × 64 · f64	11608	11.608
Matrix multiplication / faer 64 × 64 · 64 × 64 · f64	11187	11.187
Matrix multiplication / Candle 64 × 64 · 64 × 64 · f64	11379	11.379
Matrix multiplication / Kernix 64 × 64 · 64 × 64 · f64	17926	17.926

Recorded precision / exact latency table

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Matrix multiplication / NumPy 64 × 64 · 64 × 64 · f64	12204	12.204
Matrix multiplication / nalgebra 64 × 64 · 64 × 64 · f64	11195	11.195
Matrix multiplication / Kernix 128 × 128 · 128 × 128 · f64	83207	83.207
Matrix multiplication / faer 128 × 128 · 128 × 128 · f64	86726	86.726
Matrix multiplication / ndarray 128 × 128 · 128 × 128 · f64	121834	121.834
Matrix multiplication / nalgebra 128 × 128 · 128 × 128 · f64	120325	120.325
Matrix multiplication / Candle 128 × 128 · 128 × 128 · f64	78204	78.204
Matrix multiplication / NumPy 128 × 128 · 128 × 128 · f64	82343	82.343
Matrix multiplication / Kernix 256 × 256 · 256 × 256 · f64	647477	647.477
Matrix multiplication / NumPy 256 × 256 · 256 × 256 · f64	668632	668.632

Recorded precision / exact latency table

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Matrix multiplication / Candle 256 × 256 · 256 × 256 · f64	603518	603.518
Matrix multiplication / faer 256 × 256 · 256 × 256 · f64	758898	758.898
Matrix multiplication / nalgebra 256 × 256 · 256 × 256 · f64	680500	680.5
Matrix multiplication / ndarray 256 × 256 · 256 × 256 · f64	685012	685.012
Matrix multiplication / nalgebra 64 × 64 · 64 × 64 · f32	6239	6.239
Matrix multiplication / Candle 64 × 64 · 64 × 64 · f32	5767	5.767
Matrix multiplication / Kernix 64 × 64 · 64 × 64 · f32	5081	5.081
Matrix multiplication / faer 64 × 64 · 64 × 64 · f32	5624	5.624
Matrix multiplication / ndarray 64 × 64 · 64 × 64 · f32	6127	6.127
Matrix multiplication / NumPy 64 × 64 · 64 × 64 · f32	7539	7.539

Recorded precision / exact latency table

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Matrix multiplication / faer 128 × 128 · 128 × 128 · f32	50973	50.973
Matrix multiplication / NumPy 128 × 128 · 128 × 128 · f32	40227	40.227
Matrix multiplication / Candle 128 × 128 · 128 × 128 · f32	39756	39.756
Matrix multiplication / ndarray 128 × 128 · 128 × 128 · f32	52083	52.083
Matrix multiplication / Kernix 128 × 128 · 128 × 128 · f32	38634	38.634
Matrix multiplication / nalgebra 128 × 128 · 128 × 128 · f32	50480	50.48
Matrix multiplication / faer 256 × 256 · 256 × 256 · f32	343122	343.122
Matrix multiplication / Kernix 256 × 256 · 256 × 256 · f32	294932	294.932
Matrix multiplication / Candle 256 × 256 · 256 × 256 · f32	304379	304.379
Matrix multiplication / nalgebra 256 × 256 · 256 × 256 · f32	412324	412.324

Recorded precision / exact latency table

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Matrix multiplication / ndarray 256 × 256 · 256 × 256 · f32	410973	410.973
Matrix multiplication / NumPy 256 × 256 · 256 × 256 · f32	301429	301.429
Batched multiplication / Candle 4 × 16 × 32 · 1 × 32 × 16 · f64	4315	4.315
Batched multiplication / NumPy 4 × 16 × 32 · 1 × 32 × 16 · f64	3491	3.491
Batched multiplication / Kernix 4 × 16 × 32 · 1 × 32 × 16 · f64	2073	2.073
Matmul · 2 workers · reused output / Kernix 64 × 64 · 64 × 64 · f64	10422	10.422
Matmul · 2 workers · reused output / Kernix 128 × 128 · 128 × 128 · f64	105080	105.08
Matmul · 2 workers · reused output / Kernix 256 × 256 · 256 × 256 · f64	423627	423.627
Matmul · 2 workers · reused output / Kernix 128 × 128 · 128 × 128 · f32	29303	29.303
Column-pivoted QR / Kernix 96 × 32 · f64	58492	58.492

Recorded precision / exact latency table

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Column-pivoted QR / faer 96 × 32 · f64	78191	78.191
Column-pivoted QR / nalgebra 96 × 32 · f64	79440	79.44
Column-pivoted QR / Kernix 256 × 64 · f64	599809	599.809
Column-pivoted QR / faer 256 × 64 · f64	485354	485.354
Column-pivoted QR / nalgebra 256 × 64 · f64	843150	843.15
Minimum-norm least squares / Kernix 64 × 32, rank 16, 4 RHS · f64	51783	51.783
Minimum-norm least squares / nalgebra 64 × 32, rank 16, 4 RHS · f64	78173	78.173
Minimum-norm least squares / NumPy 64 × 32, rank 16, 4 RHS · f64	71772	71.772
Minimum-norm least squares / nalgebra 32 × 64, rank 32, 4 RHS · f64	142115	142.115
Minimum-norm least squares / Kernix 32 × 64, rank 32, 4 RHS · f64	81587	81.587

Recorded precision / exact latency table

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Minimum-norm least squares / NumPy 32 × 64, rank 32, 4 RHS · f64	129445	129.445
Thin QR / nalgebra 96 × 32 · f64	40857	40.857
Thin QR / NumPy 96 × 32 · f64	43608	43.608
Thin QR / Kernix 96 × 32 · f64	47578	47.578
Thin QR / faer 96 × 32 · f64	69889	69.889
Thin QR / faer 256 × 64 · f64	437890	437.89
Thin QR / Kernix 256 × 64 · f64	531300	531.3
Thin QR / NumPy 256 × 64 · f64	374776	374.776
Thin QR / nalgebra 256 × 64 · f64	395774	395.774
Cholesky factorization / NumPy 64 × 64 · f64	11778	11.778

Recorded precision / exact latency table

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Cholesky factorization / Kernix 64 × 64 · f64	21573	21.573
Cholesky factorization / faer 64 × 64 · f64	8936	8.936
Cholesky factorization / nalgebra 64 × 64 · f64	15131	15.131
Cholesky factorization / nalgebra 128 × 128 · f64	111245	111.245
Cholesky factorization / NumPy 128 × 128 · f64	74878	74.878
Cholesky factorization / faer 128 × 128 · f64	39611	39.611
Cholesky factorization / Kernix 128 × 128 · f64	120034	120.034
Cholesky factorization / nalgebra 256 × 256 · f64	752003	752.003
Cholesky factorization / faer 256 × 256 · f64	206834	206.834
Cholesky factorization / Kernix 256 × 256 · f64	685247	685.247

Recorded precision / exact latency table

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Cholesky factorization / NumPy 256 × 256 · f64	545514	545.514
Shared-variable backward / Candle sum(X * X), 64 × 32; prepared graph · f64	3961	3.961
Shared-variable backward / Kernix sum(X * X), 64 × 32; prepared graph · f64	6488	6.488
Record + backward / Candle sum(A · W), 32 × 16 · 16 × 8 · f64	4364	4.364
Record + backward / Kernix sum(A · W), 32 × 16 · 16 × 8 · f64	2518	2.518
Model inference / Candle batched normalized/sliced MLP · f64	3880	3.88
Model inference / Kernix batched normalized/sliced MLP · f64	2572	2.572
Model recording + forward / Candle batched normalized/sliced MLP · f64	5435	5.435
Model recording + forward / Kernix batched normalized/sliced MLP · f64	2926	2.926
Model backward / Kernix prepared graph → fresh gradients · f64	3311	3.311

Recorded precision / exact latency table

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Model backward / Candle prepared graph → fresh gradients · f64	14863	14.863
Parameter update / Kernix transactional SGD, fixed gradients · f64	585	0.585
Coalesced tensor contraction / Kernix 16 × 32 × 8 · 32 × 8 × 16 · f64	691795	691.795
Coalesced tensor contraction / NumPy 16 × 32 × 8 · 32 × 8 × 16 · f64	12340	12.34
Row softmax / Candle 256 × 64 · f64	93798	93.798
Row softmax / ndarray 256 × 64 · f64	85808	85.808
Row softmax / Kernix 256 × 64 · f64	93662	93.662
Row softmax / NumPy 256 × 64 · f64	110451	110.451
Create zeros / Candle 64 × 64 · f32	176	0.176
Create zeros / faer 64 × 64 · f32	160	0.16

Recorded precision / exact latency table

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Create zeros / Kernix 64 × 64 · f32	156	0.156
Create zeros / ndarray 64 × 64 · f32	90	0.09
Create zeros / NumPy 64 × 64 · f32	396	0.396
Create ones / Kernix 64 × 64 · f32	205	0.205
Create ones / Candle 64 × 64 · f32	225	0.225
Create ones / NumPy 64 × 64 · f32	1295	1.295
Create ones / ndarray 64 × 64 · f32	154	0.154
Create ones / faer 64 × 64 · f32	196	0.196